# TelcoChurn360

## Customer Churn Prediction Pipeline

ML Pipeline Summary Report

June 2026

## 1. Dataset Overview

The pipeline ingested a telecom customer dataset comprising 7,033 records across 22 features, with no duplicate or missing values at initial load.

| Metric | Value | Notes |
|---|---|---|
| Total Rows | **7,033** | After initial load |
| Total Columns | **22** | Pre-cleaning |
| Clean Rows | **7,011** | After deduplication + NaN fix |
| Final Features | **35** | Post-encoding |
| Duplicates Removed | **22** | Dropped in cleaning step |

## 2. TARGET VARIABLE — CHURN DISTRIBUTION

The dataset is moderately imbalanced, with roughly 1 in 4 customers having churned. Stratified splitting was used to preserve this ratio across train/test sets.

| Class | Count | Share |
|---|---|---|
| No Churn | **5,164** | 73.4% |
| Churn | **1,869** | 26.6% |

## 3. KEY EDA INSIGHTS — CHURN DRIVERS

Exploratory analysis revealed several strong churn signals across contract type, internet service, payment method, and customer demographics.

| Category | Segment | Churn Rate |
|---|---|---|
| **Contract** | Month-to-month | **42.7% ▲** |
| | One Year | 11.3% |
| | Two Year | 2.8% ▼ |
| **Internet Service** | Fiber Optic | **41.9% ▲** |
| | DSL | 19.0% |
| | No Internet | 7.4% ▼ |
| **Payment Method** | Electronic Check | **45.3% ▲** |
| | Credit Card (auto) | 15.3% |
| **Senior Citizen** | Yes | **41.7% ▲** |
| | No | 23.6% |

## 4. FEATURE ENGINEERING

Five new features were engineered to capture behavioral signals beyond raw columns:

- Avg_Monthly_Spend — Total Charges ÷ Tenure (spend consistency)
- Addon_Count — Number of active add-on services
- Is_New_Customer — Flag for tenure ≤ 6 months
- High_Bill — Flag for monthly charges above median
- No_Addons — Flag for customers with zero add-on services

# 5. MODEL TRAINING & EVALUATION

Three classifiers were trained with 5-Fold Stratified Cross-Validation on an 80/20 train-test split (stratified by churn label). AUC-ROC was the primary evaluation metric.

### 5.1 Cross-Validation AUC (5-Fold)

| Model | Mean AUC | Std AUC |
|---|---|---|
| Logistic Regression | **0.8600** | ±0.0202 |
| Gradient Boosting | **0.8622** | ±0.0165 |
| Random Forest | **0.8410** | ±0.0127 |

### 5.2 Test Set Performance

| Model | Test AUC | Accuracy | Churn Precision | Churn Recall |
|---|---|---|---|---|
| **Logistic Regression** □ | **0.8536** | 81% | 67% | 55% |
| Gradient Boosting | **0.8503** | 80% | 65% | 52% |
| Random Forest | **0.8299** | 79% | 65% | 50% |

# 6. PIPELINE ARTIFACTS

The following serialized files were generated at the end of the pipeline run:

- churn_model.pkl — Best trained model (Logistic Regression)
- feature_columns.pkl — Final feature list (35 columns)
- feature_importance.pkl — Feature importance scores
- label_encoders.pkl — Binary label encoders
- model_metrics.pkl — Evaluation metrics dictionary
- scaler.pkl — StandardScaler fit on training data
- sample_input.pkl — Sample input row for inference testing